



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

REVIEW TEST

27-06-2021

JEE(Main + Advance) : Nurture Course (PHASE : I,I(B),II(A) & NURTURE (LIVE)

Test Type : REVIEW TEST

Test Pattern : JEE (Advanced)

TEST DATE : 27-06-2021

ANSWER KEY

PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,B,D	A,C	B,C	A,C	A,B,C	A,B,C,D	A,B,C
SEC-2	Q.	1	2	3	4	5	6	
	A.	B	D	A	C	B	A	
SEC-3	Q.	1	2	3	4	5		
	A.	1	9	5	5	2		

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,B,C	C	A,B,D	A,B,C,D	B,C,D	D	B,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	B	C	B	D	A	A	
SEC-3	Q.	1	2	3	4	5		
	A.	1	3	3	0	3		

MATHEMATICS

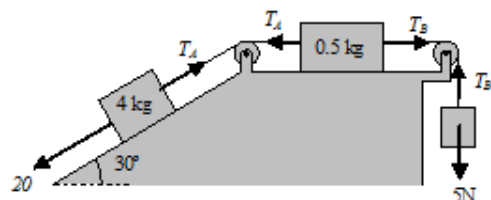
SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,C	A,C,D	A,B,C,D	B,D	A,C,D	A,B,C	B,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	D	D	B	A	C	B	
SEC-3	Q.	1	2	3	4	5		
	A.	4	9	6	0	9		

HINT – SHEET

PHYSICS

SEC-1

3. Ans (B,C)



$$(A) a = \frac{\text{Net pulling force}}{\text{total mass}} = \frac{20 - 5}{4 + 0.5 + 0.5} = 3 \text{ m/s}^2$$

(B) Applying NLM on block of mass 4 kg ;

$$20 - T_A = 4 \times 3 \Rightarrow T_A = 8 \text{ N}$$

(C) Applying NLM on block of mass 0.5 kg on top

$$T_A - T_B = 0.5 \times 3 \Rightarrow T_B = 8 - 1.5 = 6.5 \text{ N}$$

(D) Difference in tensions = 1.5 N

4. Ans (A,C)

$$2mg - T = 2ma \quad a = \frac{g}{3}$$

$$T - mg = ma$$

Since acceleration is in opposite direction

hence

$$a_{\text{rel}} = \frac{2g}{3}$$

Work by tension of a system = 0

Heavy mass moving down, Hence G.P.E

decreases.

5. Ans (A,B,C)

$$y = 10x - 2x^2$$

$$\frac{dy}{dt} = 10 \frac{dx}{dt} - 4x \frac{dx}{dt}$$

$$v_y = 10 v_x - 4x v_x$$

at highest point $v_y = 0$

$$\text{So } 10 - 4x = 0$$

$$x = 2.5$$

So range is $2.5 \times 2 = 5\text{m}$ at $x = 2.5\text{km}$ particle

is at it's maximum height

$$y = 10 \times \frac{5}{2} - 2 \times \frac{25}{4}$$

$$= \frac{50}{2} - \frac{50}{4}$$

$$= 25 - 12.5$$

$$= 12.5 \text{ m}$$

$$\text{at } x = 2.25$$

$$v_y = 10 v_x - 4 \times 2.25 v_x$$

$$v_y = 10 v_x - 9v_x$$

$$v_y = v_x$$

$$\frac{v_y}{v_x} = 1 = \tan 45$$

6. Ans (A,B,C,D)

$$a = \frac{40}{8} = 5\text{m/s}^2$$

$$T = \frac{40(x)}{100} \text{ N}$$

PHYSICS

SEC-2

4. Ans (C)

$$12 \text{ m/s}$$

$$8 \text{ m/s}^2$$

$$6 \text{ m/s}^2$$

$$u_y = 12, u_x = 0$$

$$a_y = -6, a_x = 8$$

$$t = 1, y = 12 \times 1 - \frac{1}{2} \times 6 \times 1^2 = 9$$

$$x = \frac{1}{2} \times 8 \times 1^2 = 4$$

$$v_x = 8 \times 1 = 8$$

$$v_y = 12 - 6 \times 1 = 6$$

$$v = \sqrt{6^2 + 8^2} = 10$$

5. Ans (B)

$$v_y = 12 - 6t = 0$$

$$t = 2 \text{ sec}$$

$$x = 16 \text{ m}$$

$$y = 12 \text{ m/s}$$

$$v_x = 8 \times 2 = 16 \text{ m/s}, v_y = 0, v = 16$$

6. Ans (A)

$$-12 = 12 - 6t \Rightarrow t = 4 \text{ sec}$$

$$x = 64 \text{ m}$$

$$y = 0$$

$$v_x = 32 \text{ m/s}$$

$$v_y = -12 \text{ m/s}$$

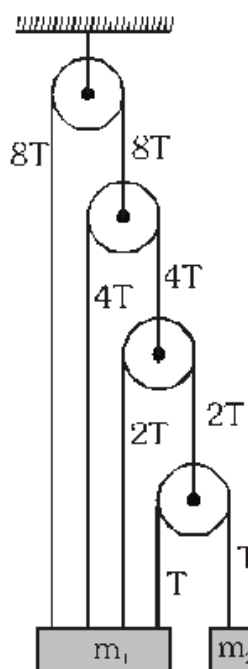
PHYSICS

SEC-3

1. Ans (1)

$$T = m_1 g \text{ \& } 8T + 4T + 2T + T = m_2 g \Rightarrow 15T = m_2 g$$

$$\Rightarrow 15 m_1 g = m_2 g \Rightarrow m_2 = 15 m_1$$



2. Ans (9)

$$\text{For equilibrium } mg \sin \theta = F \cos \theta \Rightarrow F = mg \tan \theta$$

3. **Ans (5)**

Acceleration $a =$

$$\frac{\text{Force}}{\text{total mass}} = \frac{6}{1+2} = 2 \text{ m/s}^2$$

NLM on half portion of rope $6 - T = \frac{1}{2} \times 2 \Rightarrow$

$$T = 5 \text{ N}$$

4. **Ans (5)**

For coin w.r.t. lift

$$u = 0; a = -15 \text{ m/s}; s = -7.5 \text{ m}$$

$$s = ut + \frac{1}{2}at^2$$

$$-7.5 = 0 + \frac{1}{2}(-15)t^2$$

$$t = \frac{15}{15} = 1$$

for coin w.r.t. ground

$$u = 10 \text{ m/s}; a = -g = -10 \text{ m/s}^2$$

$$s = ut + \frac{1}{2}at^2$$

$$s = 10 \times 1 + \frac{1}{2}(-10)(1)^2 = 5 \text{ m}$$

Positive so in upward direction.

5. **Ans (2)**

$$(v_b)_x = 4 - 5 \sin 37^\circ = 1 \text{ m/s}$$

$$(v_b)_y = 5 \cos 37^\circ = 4 \text{ m/s}$$

time to cross river =

$$\frac{s_y}{(v_b)_y} = \frac{8000}{4} = 2000 \text{ sec}$$

$$\text{drift } s_x = (v_b)_x t = 1 \times 2000 = 2 \text{ km}$$

CHEMISTRY

SEC-1

2. **Ans (C)**

Ans. (C)

5. **Ans (B,C,D)**

$$(A) H_2 > He$$

More Vander waal force in case of H_2 due to its large size as compare to He.

More attraction μ More boiling point

$$(B) H_2 < D_2$$

More molar mass

As molar mass increases it becomes difficult to boil the substance.

Molar mass increases μ Boiling point increases.

$$(C) H_2O < D_2O$$

Slightly above* Extant of H-bonding is more in D_2O

* Molecular weight of D_2O is more compare to H_2O

$$(D) NH_3 < SbH_3$$

Order of boiling point in case of 15th group hydrides.

$$BiH_3 > SbH_3 > NH_3 > AsH_3 > PH_3$$

* General trend is due to increase in molecular mass.

* But due to H-bonding there are some changes in case of boiling point of NH_3 as it becomes higher compare to AsH_3 .

MATHEMATICS

SEC-1

1. **Ans (A,C)**

$$5^{\log_5(x^2-5x+7)} = x - 1$$

$$x^2 - 5x + 7 - x + 1 = 0$$

$$x^2 - 6x + 8 = 0$$

$$(x - 4)(x - 2) = 0$$

$$x = 4 \text{ or } x = 2$$

2. **Ans (A,C,D)**

$$(pqr)^{\frac{1}{3}} \geq \frac{(p+q+r)}{3} \Rightarrow GM \geq AM$$

possible when $AM = GM \Rightarrow p = q = r$

$$3p + 4q + 5r = 12 \Rightarrow p = q = r = 1$$

3. Ans (A,B,C,D)

$$a^3 + b^3 = (a+b)^3 \Rightarrow a=0 \text{ or } b=0 \text{ or } a+b=0$$

$$\log_{10}x + 2 = 0 \text{ or } \log_{10}x - 1 = 0 \text{ or } 2\log_{10}x + 1 = 0$$

$$x = 10^{-2} \quad x = 10 \quad x = 10^{-1/2}$$

4. Ans (B,D)

$$A = 3^{400}$$

$$\log_{10}A = 400\log_{10}3 \\ = 190.84$$

$$m = 190 + 1 = 191$$

$$B = 2^{-200}$$

$$\log_{10}B = -200\log_{10}2 \\ = -60.2$$

$$\therefore n = |-61 + 1| = 60$$

5. Ans (A,C,D)

$$T_n = (2^n) + (n^2 + 1)$$

$$S_n = \sum_{n=1}^n T_n$$

$$= \sum_{n=1}^n (2^n) + \sum_{n=1}^n (n^2) + \sum_{n=1}^n (1)$$

$$= \frac{2(2^n - 1)}{2 - 1} + \frac{n(n+1)(2n+1)}{6} + n$$

$$= 2^{n+1} + \frac{n^3}{3} + \frac{n^2}{2} + \frac{7n}{6} - 2$$

$$A = 1, B = \frac{1}{3}, C = \frac{1}{2},$$

$$D = \frac{7}{6}, E = -2$$

6. Ans (A,B,C)

$$S = 1 + \left(1 + \frac{1}{2}\right) \left(\frac{1}{3}\right) + \left(1 + \frac{1}{2} + \frac{1}{2^2}\right) \left(\frac{1}{3^2}\right) + \dots \\ \dots(1)$$

$$\frac{S}{3} = \frac{1}{3} + \left(1 + \frac{1}{2}\right) \left(\frac{1}{3^2}\right) + \dots \\ \dots(2)$$

$$\text{Equation (1) - (2)}$$

$$\frac{2S}{3} = 1 + \frac{1}{6} + \frac{1}{6^2} + \dots = \frac{6}{5}$$

$$S = \frac{18}{10} = \frac{9}{5}$$

7. Ans (B,D)

Given that

$$\text{Sum} = 4 \text{ and } T_2 = \frac{3}{4}$$

$$\text{We know } S_\infty = \frac{a}{1-r}, T_2 = ar = \frac{3}{4}, a = \frac{3}{4r}$$

$$S_\infty = \frac{\frac{3}{4r}}{1-r} = 4$$

$$\frac{3}{4r} = 4 - 4r$$

$$3 = 16r - 16r^2$$

$$16r^2 - 16r + 3 = 0$$

$$16r^2 - 12r - 4r + 3 = 0$$

$$4r(4r-3) - 1(4r-3) = 0$$

$$(4r-1)(4r-3) = 0$$

$$r = \frac{1}{4}, \frac{3}{4}$$

$$\text{Now } a = \frac{3}{4r}$$

$$\text{When } r = \frac{1}{4} \Rightarrow \boxed{a=3}$$

$$\text{When } r = \frac{3}{4} \Rightarrow \boxed{a=1}$$

MATHEMATICS

SEC-2

1. Ans (D)

$$(I) x^2 - 3x + 2 = 0 \quad x = 1, 2 \\ \text{sum} = 3, \text{product} = 2$$

$$(II) \log_3((x^2 - 2)(x - 1)) = \log_3 2 \\ x^3 - x^2 - 2x + 2 = 2 \\ x(x^2 - x - 2) = 0 \\ x = 0, x = 2, x = -1 \\ x = 2 \text{ is only solution} \\ \text{sum} = 2, \text{product} = 2$$

$$(III) x^2 - 4 = 0 \Rightarrow x = 2, -2 \\ |x - 2| = 1 \Rightarrow x = 3 \text{ or } x = 1 \\ x = 2 \text{ is rejected} \\ \text{sum} = -2 + 3 + 1 = 2, \\ \text{product} = (-2)(3)(1) = -6$$

$$(IV) 2^{(\log_2 x)^2} + x^{\log_2 x} = 32 \\ x^{\log_2 x} = 16 \\ (\log_2 x)^2 = 4$$

$$x = 2^2 \text{ \& } x = 2^{-2} = \frac{1}{4}$$

$$\text{sum} = 4 + \frac{1}{4} = \frac{17}{4} \\ \text{product} = 1$$

2. **Ans (D)**

(I) $x^2 - 3x + 2 = 0$ $x = 1, 2$

sum = 3, product = 2

(II) $\log_3((x^2 - 2)(x - 1)) = \log_3 2$

$x^3 - x^2 - 2x + 2 = 2$

$x(x^2 - x - 2) = 0$

$x = 0, x = 2, x = -1$

$x = 2$ is only solution

sum = 2, product = 2

(III) $x^2 - 4 = 0 \Rightarrow x = 2, -2$

$|x - 2| = 1 \Rightarrow x = 3$ or $x = 1$

$x = 2$ is rejected

sum = $-2 + 3 + 1 = 2$,

product = $(-2)(3)(1) = -6$

(IV) $2^{(\log_2 x)^2} + x^{\log_2 x} = 32$

$x^{\log_2 x} = 16$

$(\log_2 x)^2 = 4$

$x = 2^2$ & $x = 2^{-2} = \frac{1}{4}$

sum = $4 + \frac{1}{4} = \frac{17}{4}$

product = 1

3. **Ans (B)**

(I) $x^2 - 3x + 2 = 0$ $x = 1, 2$

sum = 3, product = 2

(II) $\log_3((x^2 - 2)(x - 1)) = \log_3 2$

$x^3 - x^2 - 2x + 2 = 2$

$x(x^2 - x - 2) = 0$

$x = 0, x = 2, x = -1$

$x = 2$ is only solution

sum = 2, product = 2

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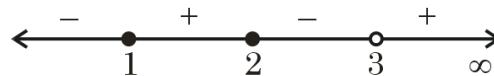
$x = 2^2$ & $x = 2^{-2} = \frac{1}{4}$

sum = $4 + \frac{1}{4} = \frac{17}{4}$

product = 1

4. **Ans (A)**

(I) $\frac{(x-1)(x-2)}{(x-3)} \geq 0$



$x \in [1, 2] \cup (3, \infty)$

Infinitely many integral values of x

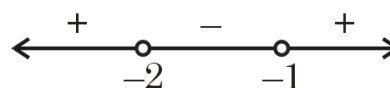
(I) (ii) (S) is correct.

(II) $\frac{x}{x+1} > 2$

$\frac{x}{x+1} - 2 > 0$

$\frac{-x-2}{x+1} > 0$

$\frac{x+2}{x+1} < 0$



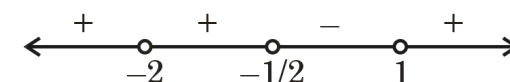
$x \in (-2, -1)$

Number of integral values of x is zero

(II) (iii) (P) is correct.

(III) $\frac{x^2 + 4x + 4}{2x^2 - x - 1} < 0$

$\frac{(x+2)^2}{(x-1)(2x+1)} < 0$



$x \in \left(-\frac{1}{2}, 1\right)$

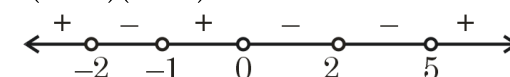
Number of integral values of $x = 1$

(III) (iv) (Q) is correct.

(IV) $\frac{x^3(x-2)(5-x)}{(x^2-4)(x+1)} > 0$

$\frac{x^3(x-2)(x-5)}{(x-2)(x+2)(x+1)} < 0$

$\frac{x^3(x-5)}{(x+2)(x+1)} < 0, x \neq 2$



$x \in (-2, -1) \cup (0, 5) - \{2\}$

Number of integral values of $x = 3$

$(x = 1, 3, 4)$

(IV) (i) (R) is correct.

5. Ans (C)

$$(I) \frac{(x-1)(x-2)}{(x-3)} \geq 0$$



$$x \in [1, 2] \cup (3, \infty)$$

Infinitely many integral values of x

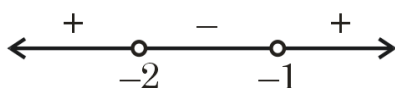
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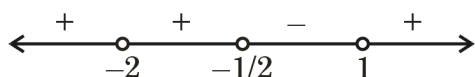
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Number of integral values of x is zero

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$$\frac{(x+2)^2}{(x-1)(2x+1)} < 0$$



$$x \in \left(-\frac{1}{2}, 1\right)$$

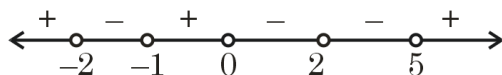
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$$\frac{x^3(x-2)(x-5)}{(x-2)(x+2)(x+1)} < 0$$

$$\frac{x^3(x-5)}{(x+2)(x+1)} < 0, \quad x \neq 2$$



$$x \in (-2, -1) \cup (0, 5) - \{2\}$$

Number of integral values of $x = 3$

$$(x = 1, 3, 4)$$

(IV) (i) (R) is correct.

6. Ans (B)

$$(I) \frac{(x-1)(x-2)}{(x-3)} \geq 0$$



$$x \in [1, 2] \cup (3, \infty)$$

Infinitely many integral values of x

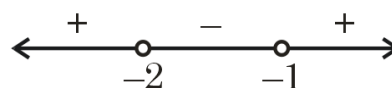
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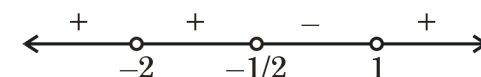
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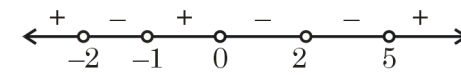
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$$\frac{x^3(x-5)}{(x+2)(x+1)} < 0, \quad x \neq 2$$



$$x \in (-2, -1) \cup (0, 5) - \{2\}$$

Number of integral values of $x = 3$

$$(x = 1, 3, 4)$$

(IV) (i) (R) is correct.

MATHEMATICS
SEC-3

 1. **Ans (4)**

$$\frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{b} + \frac{1}{b} + \frac{1}{c} + \frac{1}{c} + \frac{1}{c} + \frac{1}{c}}{8} \geq \left(\frac{1}{ab^3c^4} \right)^{1/8} \geq \frac{1}{2}$$

$$\Rightarrow \frac{1}{a} + \frac{3}{b} + \frac{4}{c} \geq \frac{8}{2} \text{ hence 4.}$$

 2. **Ans (9)**

$$S = 3 \cdot 3^{2/2} \cdot 3^{3/4} \cdot 3^{4/8} \dots \infty$$

$$S = 3^1 + \frac{2}{2} + \frac{3}{4} + \frac{4}{8} + \dots \infty$$

$$S' = 1 + \frac{2}{2} + \frac{3}{4} + \frac{4}{8} + \dots \infty \dots (1)$$

$$\frac{S'}{2} = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \dots \infty \dots (2)$$

subtract (2) from (1)

$$\frac{S'}{2} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \infty$$

$$S' = 4$$

$$S = 3^4 = 81$$

 3. **Ans (6)**

$$a = 10, b = 28, n = 11$$

$$d = \frac{b-a}{n+1} = \frac{3}{2}$$

$$A_k = a + kd$$

$$A_8 - A_4 = 4d = 6$$

 4. **Ans (0)**

LHS = Positive

RHS = Negative

 $\therefore$ No solution

 5. **Ans (9)**

$$\log_{\sqrt{2}} \left(\text{antilog}_{128} \left(\frac{3}{7} \right) \right) = \log_{\sqrt{2}} (128)^{\frac{3}{7}} = 6$$

$$\log_2 \left(\text{antilog}_{128} \left(\frac{3}{7} \right) \right) = \log_2 (128)^{\frac{3}{7}} = 3$$